

Project - Flight Punctuality Analytics

1. Purpose

The project aims to implement the modern data stack to solve a real-world problem. The focus of the project is to build the modern data stack according to the business requirements independently as a team of data engineers.

2. Scenario

Suppose your group work as the data engineering team for the hypothetical airport authority in Sweden. In the coming sprint of two weeks, your team is asked to produce a proof of concept (PoC) for a modern data stack that assists the analysis of flight punctuality. Your team have already met the business stakeholders and below are the business requirements that you have got:

3. Business requirements

You should first work with the mandatory requirements and then with the additional ones.

Mandatory requirements

- build a modern data stack with **dlt**, **dbt core**, **Snowflake**, **streamlit** and **dagsters**, which means that your data pipeline involves data loading, data transformation, cloud data warehouse, BI visualization and orchestration
- the primary interest of the business stakeholders is to analyse the numbers of delayed arrivals at and departures from Swedish airports. They would also like to analyse the delayed arrivals and departures by airlines, airports and dates
- create a dimensional model for the transformed data according to the primary interest of the business stakeholders. Use **dbdiagram** for this part
- create a dashboard that business stakeholders can use for their analysis
- use these available data sources:
 - Swedavia FlightInfo API. Sign up for the access and obtain an API key according to the [instruction](#). Check out the API documentation [here](#). Use data from both the endpoints **arrivals** and **departures**. You should extract data covering the past three days (including the date when you extract data) and all available airports in Sweden
 - Airline operator info from [Wikipedia](#)
 - Airport info from [Wikipedia](#)
- follow Snowflake access control's best practices when setting up your cloud data warehouse. You should develop within THE SAME trial account of Snowflake

Additional requirements

- containerize your codes with Docker, separately for dagsters and streamlit. Test your containers locally
- deploy your containers to your Azure student account. Spin up **container app** and **web app** resources. For the group presentation, you can show the deployment in one Azure student account

4. Submission and Grading

There are group and individual submissions for this project. For both submissions, you can receive the grade of IG, G or VG. For each submission, you obtain G if you have fulfilled the mandatory business requirements and VG if you fulfilled both mandatory and additional business requirements.

In order to get G in this course, both submissions should be at least G. For VG in this course, you need to obtain VG for both submissions.

Below are details for each submission:

Group

You'll submit your shared code base in GitHub. The commit history of the code base should show what each person has done during the project. Each group will present for 10 minutes, including:

- briefly present how data flowed from upstream to downstream
- illustrate your data model for the transformed data
- demonstrate how the dashboard assists decision making of the airport authority

Include a README.md for your repo to document your project which can guide others to understand your project. Use functions and modules to well-structure your scripts.

Individual

A 5-10 minute video illustrating how to run the pipeline locally on your own computer by using the github repo of your group. For the additional requirement, you also need to show the cloud deployment in your own Azure student account.

The focus of this individual submission is that you are able to run the pipeline independently and understanding how different components of your code base contribute to your modern data stack.